

Lab: The District That Was Not Drawn

84-355 Democracy's Data | Vote dilution, Houston County, Georgia

Jonathan Cervas

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Everything here runs as given. Your job is to read what comes out and say what it means.

Before we start

Vote dilution is the claim that your vote is worth less than somebody else's. American law recognises two versions and they are not alike.

Malapportionment is districts of unequal population. *Reynolds v. Sims*, 377 U.S. 533 (1964) held the Equal Protection Clause forbids it. You settle it by counting people.

Racial vote dilution under Section 2 of the Voting Rights Act, 52 U.S.C. § 10301, is districts drawn so that a cohesive minority group, voted against by a cohesive majority, cannot elect anyone. *Thornburg v. Gingles*, 478 U.S. 30, 50–51 (1986) made a plaintiff show three things first: the group is **sufficiently large and geographically compact** to be a majority in a single-member district; it is **politically cohesive**; and the majority **votes as a bloc** usually to defeat its preferred candidate.

Last week did preconditions two and three. The RPV lab estimated whether Black and white voters preferred different candidates, and graded the estimate against records that happen to exist.

Today does precondition one, and it is a different kind of claim. The other two are about an election that happened. This one is about **a district that does not exist** — that a map *could* have been drawn in which the group would have been a majority. Nobody drew it. So we have to.

The law changed four months ago. *Louisiana v. Callais*, 608 U.S. ____ (2026), decided 29 April 2026, left the three preconditions standing and changed what satisfies them. Precondition one now carries **two** conditions. A plaintiff “cannot use race as a districting criterion” in drawing an illustrative map; and the map “must meet all the State’s legitimate districting objectives, including traditional districting criteria and the State’s specified political goals,” achieving them **“just as well”** as the State’s own map (slip op., at 29). Part 5 builds a district with a dial that optimises race and nothing else, because that was standard practice until this spring. Part 6 says which end of the dial survives the first condition — and why nothing in this lab survives the second.

For Parts 5 and 6, **pretend the enacted map does not exist.** That is the position of an expert asked whether a district *could* be drawn.

Part 1 — What you observe

```
b <- read.csv("data/blocks.csv", colClasses = c(GEOID20 = "character"))
a <- read.csv("data/adjacency.csv")
```

```
cat("census blocks:      ", nrow(b), "\n")
```

```
#> census blocks:      3269
```

```
cat("population:        ", sum(b$pop), "\n")
```

```
#> population:        163633
```

```
cat("voting-age pop:     ", sum(b$vap), "\n")
```

```
#> voting-age pop:     122118
```

```
cat("registered voters:  ", sum(b$reg), "\n\n")
```

```
#> registered voters:  49771
```

```
cat(sprintf("Black share of population:      %.2f%%\n",
            100 * sum(b$pop_black) / sum(b$pop)))
```

```
#> Black share of population:      34.54%
```

```
cat(sprintf("Black share of voting-age pop:    %.2f%%\n",
            100 * sum(b$vap_black) / sum(b$vap)))
```

```
#> Black share of voting-age pop:   32.43%
```

```
cat(sprintf("Black share of registered:        %.2f%%\n",
            100 * sum(b$reg_black) / sum(b$reg)))
```

```
#> Black share of registered:      35.56%
```

One row per **2020 census block**, the atom every districting plan is built from. Houston County's Board of Education is elected from five single-member districts under a plan enacted by the Georgia General Assembly; column `boe5` says which district each block is in.

Read those last three numbers again. The Black share of this county is one of three different things depending on which population you mean. Part 7 is about what that does.

Part 2 — The dilution you settle by subtraction

```

ag <- aggregate(cbind(pop, vap, vap_black) ~ boe5, b, sum)
ideal <- mean(ag$pop)
ag$deviation <- 100 * (ag$pop - ideal) / ideal

print(ag[, c("boe5", "pop", "deviation")], row.names = FALSE, digits = 3)

```

```

#> boe5 pop deviation
#> 1 32638 -0.271
#> 2 32579 -0.451
#> 3 33090 1.110
#> 4 32550 -0.540
#> 5 32776 0.151

```

```

cat(sprintf("\nideal district: %.0f people\n", ideal))

```

```

#>
#> ideal district: 32727 people

```

```

cat(sprintf("total spread: %.2f points\n", max(ag$deviation) - min(ag$deviation)))

```

```

#> total spread: 1.65 points

```

```

cat(sprintf("a vote in the smallest district is worth %.2f%% more than in the largest\n",
100 * (max(ag$pop) / min(ag$pop) - 1)))

```

```

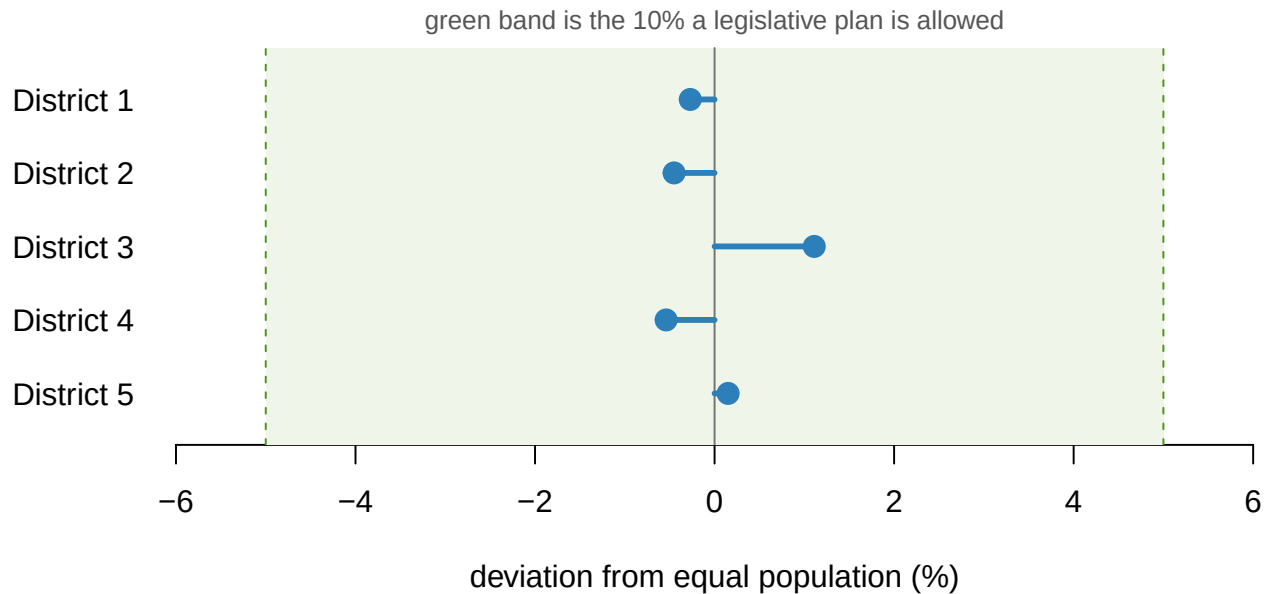
#> a vote in the smallest district is worth 1.66% more than in the largest

```

```

par(mar = c(4.4, 5.6, 2.0, 1.4))
y <- 5:1
plot(NA, xlim = c(-5.6, 5.6), ylim = c(0.5, 5.5), yaxt = "n", bty = "n", las = 1,
xlab = "deviation from equal population (%)", ylab = "")
rect(-5, 0.3, 5, 5.7, col = "#f0f5ea", border = NA)
abline(v = 0, col = "grey45"); abline(v = c(-5, 5), lty = 2, col = "#4d9221")
axis(2, at = y, labels = paste("District", ag$boe5), las = 1, tick = FALSE)
segments(0, y, ag$deviation, y, col = "#2c7fb8", lwd = 3)
points(ag$deviation, y, pch = 19, cex = 1.5, col = "#2c7fb8")
mtext("green band is the 10% a legislative plan is allowed", 3, line = 0.3,
cex = 0.82, col = "#555")

```



Brown v. Thomson, 462 U.S. 835, 842 (1983) collected the rule that a state or local plan under a **10%** maximum deviation is a minor deviation that does not by itself make out a case. **This plan uses a sixth of that allowance.** The malapportionment inquiry is over.

If it is this easy, what was it for?

Part 3 — What that rule replaced

Until 1963 Georgia nominated statewide officers under the **county unit system**: counties cast unit votes, not people — six each for the eight largest, four for the next thirty, two for the remaining hundred and twenty-one — and a candidate needed a majority of units. *Gray v. Sanders*, 372 U.S. 368 (1963) struck it down, and it is where Justice Douglas wrote, at 381, that political equality “can mean only one thing — one person, one vote.”

That rule, unchanged, on Georgia’s 2020 population:

```
cu <- read.csv("data/ga_county_units.csv", colClasses = c(fips = "character"))
cat(sprintf("people per unit vote, largest county: %8.0f\n",
            max(cu$people_per_unit)))
```

```
#> people per unit vote, largest county:    177785
```

```
cat(sprintf("people per unit vote, smallest county: %8.0f\n",
            min(cu$people_per_unit)))
```

```
#> people per unit vote, smallest county:      780
```

```
cat(sprintf("ratio:                                %8.0f to 1\n",
            max(cu$people_per_unit) / min(cu$people_per_unit)))
```

```
#> ratio:                                228 to 1
```

```
o <- cu[order(cu$people_per_unit), ]
k <- which(cumsum(o$units) >= floor(sum(cu$units) / 2) + 1)[1]
cat(sprintf("\n%d of 159 counties could cast a unit-vote majority,\n", k))
```

```
#>
#> 103 of 159 counties could cast a unit-vote majority,
```

```
cat(sprintf("and they hold %.1f%% of Georgians.\n",
            100 * sum(o$pop[seq_len(k)]) / sum(cu$pop)))
```

```
#> and they hold 15.3% of Georgians.
```

Those are the numbers *Reynolds* and *Gray* were written against. The equal-population rule worked so completely that it stopped being where the argument is.

Part 4 — The dilution you cannot settle by subtraction

The first *Gingles* precondition has two halves. Only the first is arithmetic.

```
d_vap <- sum(b$vap) / 5           # voting-age people per district
need_bl <- d_vap / 2              # Black adults a bare majority would take
have_bl <- sum(b$vap_black)
```

```
cat(sprintf("voting-age people per district: %8.0f\n", d_vap))
```

```
#> voting-age people per district:    24424
```

```
cat(sprintf("Black adults a majority needs: %8.0f\n", need_bl))
```

```
#> Black adults a majority needs:    12212
```

```
cat(sprintf("Black adults the county has: %8.0f\n", have_bl))
```

```
#> Black adults the county has:    39605
```

```
cat(sprintf("\n'sufficiently large' is satisfied %.2f times over\n",
            have_bl / need_bl))
```

```
#>
#> 'sufficiently large' is satisfied 3.24 times over
```

Bartlett v. Strickland, 556 U.S. 1 (2009), a plurality, read the precondition to require a real numerical majority — greater than 50 percent of voting-age population, at 19–20. A district the group could win only with crossover help “§ 2 does not require,” at 25.

That leaves “**geographically compact.**” The Court’s current phrase is a “**reasonably configured**” district, from *Wisconsin Legislature v. Wisconsin Elections Commission*, 595 U.S. 398, 402 (2022); *Allen v. Milligan*, 599 U.S. 1, 18 (2023) glossed it as one that “comports with traditional districting criteria.” It is a fact about where people live.

```

bs <- read.csv("data/block_shapes.csv", colClasses = c(id = "character"))
ps <- read.csv("data/plan_shapes.csv")

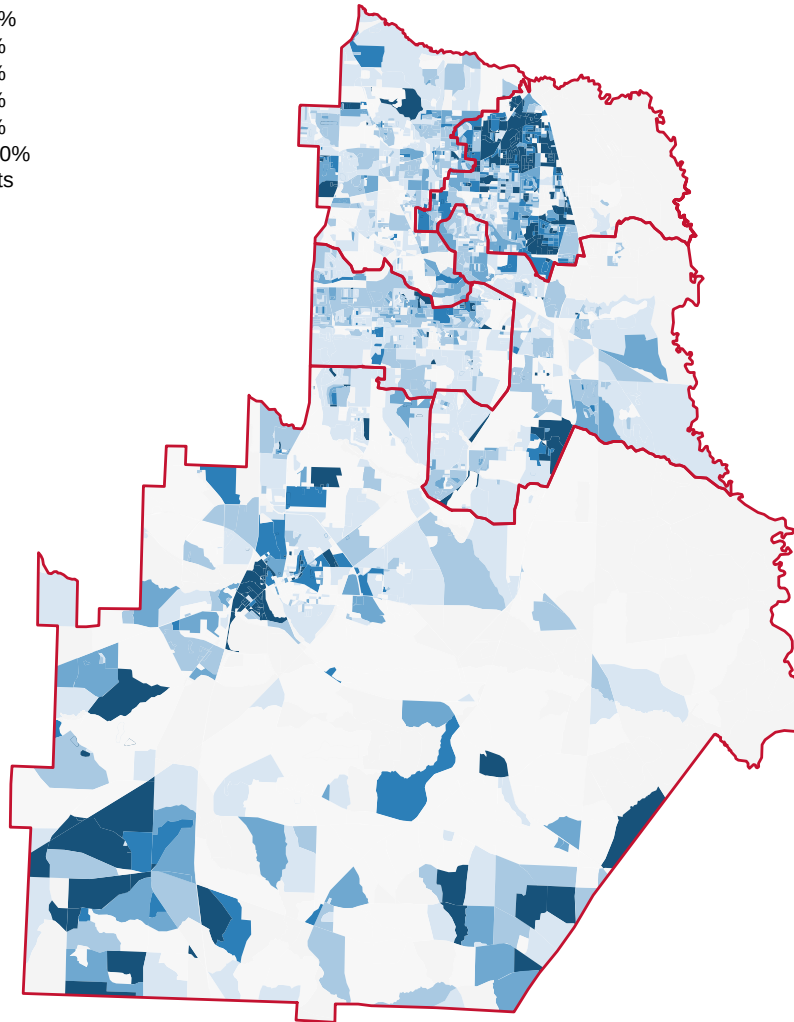
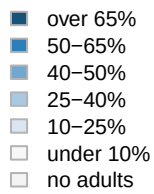
share <- ifelse(b$vap > 0, b$vap_black / b$vap, NA)
brk <- c(0, .10, .25, .40, .50, .65, 1)
col <- c("#f7f7f7", "#d9e6f2", "#a9c9e2", "#6fa8d0", "#2c7fb8", "#17527a")
fill <- ifelse(is.na(share), "#f4f4f4", col[cut(share, brk, include.lowest = TRUE)])

par(mar = c(0.4, 0.4, 2.0, 0.4))
plot(NA, xlim = range(bs$x), ylim = range(bs$y), asp = 1, axes = FALSE,
     xlab = "", ylab = "")
for (r in split(bs, interaction(bs$id, bs$part, drop = TRUE)))
  polygon(r$x, r$y, col = fill[match(r$id[1], b$GEOID20)], border = NA)
for (dd in 1:5) {
  z <- ps[ps$id == paste("Enacted BOE district", dd), ]
  for (r in split(z, z$part)) lines(r$x, r$y, col = "#C41230", lwd = 1.4)
}
legend("topleft", c("over 65%", "50-65%", "40-50%", "25-40%", "10-25%",
                    "under 10%", "no adults"),
      fill = c(rev(col), "#f4f4f4"), border = "grey70", bty = "n", cex = 0.7,
      title = "Black share of a block's adults", title.adj = 0)
mtext("red lines are the five enacted districts", 3, line = 0.4, cex = 0.82,
      col = "#555")

```

red lines are the five enacted districts

Black share of a block's adults



```
sg <- read.csv("data/segregation.csv")
print(sg, row.names = FALSE)
```

```
#>                                     quantity value
#> index of dissimilarity, Black vs non-Black adults 39.88
```

```
hi <- b$vap >= 20 & b$vap_black / pmax(b$vap, 1) > 0.5
cat(sprintf("\nblocks where Black adults are already a majority: %d of %d\n",
            sum(hi), nrow(b)))
```

```
#>
#> blocks where Black adults are already a majority: 306 of 3269
```

```
cat(sprintf("share of the county's Black adults living in them: %.1f%%\n",
            100 * sum(b$vap_black[hi]) / have_bl))
```

```
#> share of the county's Black adults living in them: 36.4%
```

The **index of dissimilarity** is the share of Black adults who would have to move blocks for the two groups to be spread identically. This is moderate segregation, not severe — and **a group can be plenty large and still be spread too evenly to be gathered into a majority anywhere.**

Part 5 — One algorithm with one dial

To find out whether the district exists, build it. Start at one block and annex neighbours until the district holds the ideal population. At each step the mapmaker looks at the **reach** nearest available blocks on the frontier and takes whichever leaves the Black share of adults highest.

```
# adjacency list: which blocks touch which. Row numbers of `b`.
nb <- split(c(a$a, a$b), c(a$b, a$a))
NB <- vector("list", nrow(b))
NB[as.integer(names(nb))] <- nb

grow <- function(seed, target, reach = Inf) {
  inn <- logical(nrow(b)); inn[seed] <- TRUE
  P <- b$pop[seed]; N <- b$vap_black[seed]; D <- b$vap[seed]
  cx <- b$x_km[seed]; cy <- b$y_km[seed]
  fr <- NB[[seed]]
  while (P < target && length(fr)) {
    if (is.finite(reach) && reach < length(fr)) {
      dd <- (b$x_km[fr] - cx)^2 + (b$y_km[fr] - cy)^2
      cand <- fr[order(dd)[seq_len(reach)]]
    } else cand <- fr
    pick <- cand[which.max((N + b$vap_black[cand]) / pmax(D + b$vap[cand], 1))]
    inn[pick] <- TRUE
    P <- P + b$pop[pick]; N <- N + b$vap_black[pick]; D <- D + b$vap[pick]
    w <- b$pop[inn] + 1
    cx <- sum(b$x_km[inn] * w) / sum(w); cy <- sum(b$y_km[inn] * w) / sum(w)
    fr <- setdiff(unique(c(fr, NB[[pick]])), which(inn))
  }
  which(inn)
}

# how Black is a set of blocks, three ways?
score <- function(i) c(
  population = 100 * sum(b$pop_black[i]) / sum(b$pop[i]),
  adults      = 100 * sum(b$vap_black[i]) / sum(b$vap[i]),
  registered  = 100 * sum(b$reg_black[i]) / sum(b$reg[i]))

seed <- which.max(b$vap_black) # the block with most Black adults
cat("seed block:", b$GE0ID20[seed], "with", b$vap_black[seed], "Black adults\n")
```

```
#> seed block: 131530211162000 with 413 Black adults
```

- **reach = 1.** One candidate is ever considered, so race never enters the *annexation rule*. The roundest district the geography allows — and after *Callais*, the only end of this dial that could be offered to a court.
- **reach = Inf.** Every frontier block is a candidate, however far out. One objective, no other constraint.

Look at the line above the bullets before you accept that. `seed` is `which.max(b$vap_black)` — the block picked *because* it holds the most Black adults. Race chose the starting point at every setting of the

dial, `reach = 1` included. Turning the dial down takes race out of how the district grows and leaves it in where the district begins. Option B is where you can test what that is worth.

```
round(rbind(`reach = 1` = score(grow(seed, ideal, 1)),
            `reach = Inf` = score(grow(seed, ideal, Inf))), 2)
```

```
#>           population adults registered
#> reach = 1         45.34  41.27         48.17
#> reach = Inf        62.09  60.00         58.04
```

A third construction abandons contiguity: take the most heavily Black blocks in order, wherever they are. Nobody could enact it. It is a ceiling.

```
o <- order(b$vap_black / pmax(b$vap, 1), b$vap_black, decreasing = TRUE)
ceiling <- o[seq_len(sum(cumsum(b$pop[o]) < ideal) + 1)]
round(score(ceiling), 2)
```

```
#> population      adults registered
#>      68.20      67.10      62.78
```

Part 6 — Turning the dial

```
tr <- read.csv("data/tradeoff.csv")
t5 <- tr[tr$districts == 5, ]
print(t5[, c("reach", "vap_black_pct", "reg_black_pct", "pp", "reock")],
      row.names = FALSE, digits = 4)
```

```
#> reach vap_black_pct reg_black_pct pp reock
#>    1         41.27         48.17 0.324 0.766
#>    2         41.14         47.85 0.251 0.762
#>    3         43.46         48.83 0.273 0.772
#>    4         43.58         48.85 0.202 0.769
#>    6         42.83         48.92 0.174 0.727
#>    8         44.32         49.61 0.161 0.759
#>   12         45.19         50.23 0.118 0.756
#>   16         45.11         49.86 0.086 0.689
#>   24         47.69         52.27 0.074 0.599
#>   40         51.07         55.57 0.036 0.336
#>   60         53.87         57.26 0.029 0.315
#>  100         55.18         58.86 0.023 0.309
#>  200         42.68         41.84 0.006 0.248
#>  500         60.00         58.04 0.008 0.284
#>  Inf         60.00         58.04 0.008 0.284
```

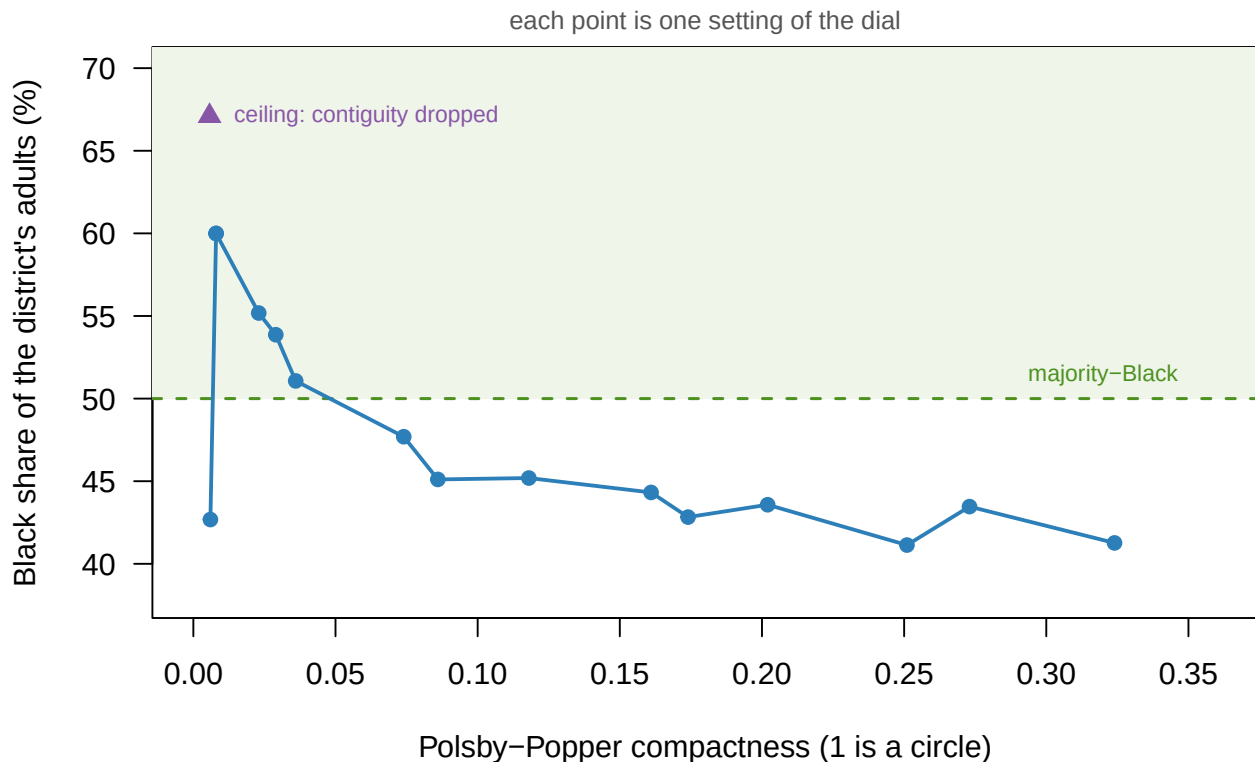
Polsby-Popper is $4\pi A / P^2$: one for a circle, near zero for anything with a long ragged edge. **Reock** is the district's area over the area of the smallest circle containing it. Both are computed on the unsimplified geometry by `data/build-data.R`, because base R cannot do it.

```

d <- read.csv("data/districts.csv")
e4 <- d[d$plan == "Enacted BOE district 4", ]
cl <- d[d$plan == "Ceiling, 5 districts", ]

par(mar = c(4.4, 4.6, 2.0, 1.2))
plot(NA, xlim = c(0, 0.36), ylim = c(38, 70), las = 1,
     xlab = "Polsby-Popper compactness (1 is a circle)",
     ylab = "Black share of the district's adults (%)")
rect(-1, 50, 1, 100, col = "#f0f5ea", border = NA)
abline(h = 50, lty = 2, col = "#4d9221", lwd = 1.6)
z <- t5[order(t5$pp), ]
lines(z$pp, z$vap_black_pct, col = "#2c7fb8", lwd = 2)
points(t5$pp, t5$vap_black_pct, pch = 19, col = "#2c7fb8")
points(cl$pp, cl$vap_black_pct, pch = 17, cex = 1.3, col = "#8856a7")
text(cl$pp, cl$vap_black_pct, "ceiling: contiguity dropped", pos = 4,
     cex = 0.72, col = "#8856a7")
text(0.355, 51.4, "majority-Black ", pos = 2, cex = 0.74, col = "#4d9221")
mtext("each point is one setting of the dial", 3, line = 0.3, cex = 0.82,
     col = "#555")

```



The dial buys Black share and pays in compactness, at a brutal rate. At reach = 1 the district is a respectable shape and nowhere near a majority. The first setting that clears 50% does so with a Polsby-Popper roughly nine times worse.

One setting is out of order — the district at reach = 200 drops below several tighter settings. **A greedy algorithm is not guaranteed to be monotone in its own parameter.** The curve an expert report shows you is a property of the search, not of the county.

And read it with the calendar in mind. Every point except the leftmost was produced by letting race decide which block to annex. **After *Callais* a plaintiff “cannot use race as a districting criterion,”** which

leaves reach = 1.

That is the generous reading. The second condition asks more than this chart can answer: the illustrative map must meet **all** the State’s legitimate objectives — traditional criteria and “the State’s specified political goals” — “**just as well**” as the enacted map. Nothing here is a complete plan, nothing here knows what Georgia’s objectives were, and no election result is in the folder. **On the current standard the whole chart is inadmissible, not just its right-hand side.**

The honest write-up: *in Houston County a majority-Black district can be constructed only by abandoning compactness.*

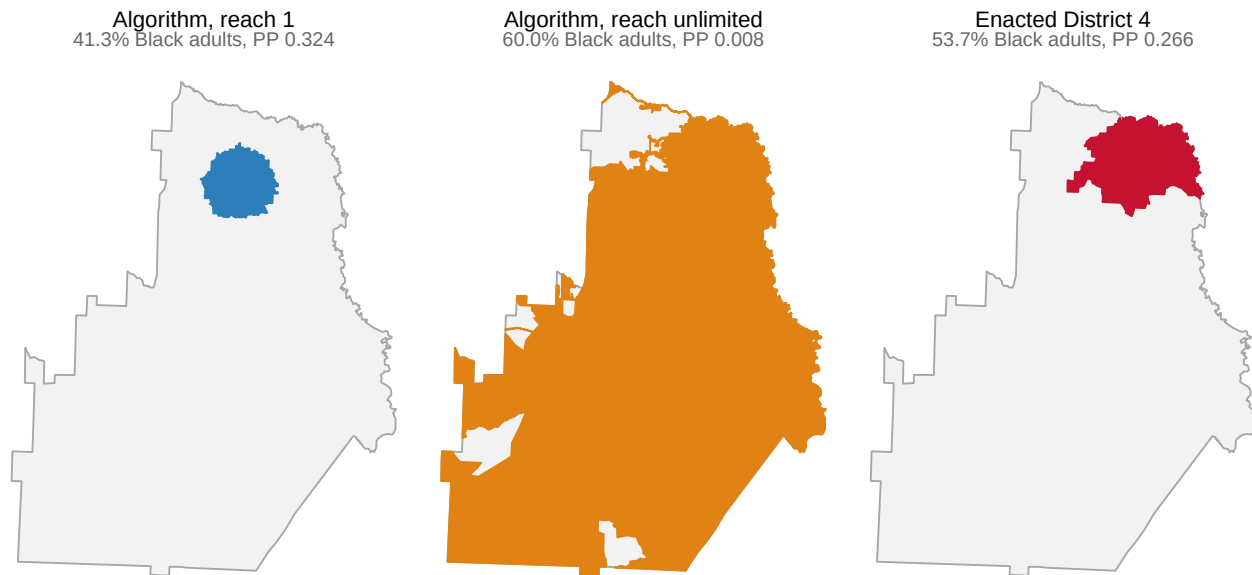
That is wrong.

Part 7 — The map already exists

```
print(d[, c("plan", "pop", "vap_black_pct", "reg_black_pct", "pp", "reock")],
      row.names = FALSE, digits = 4)
```

```
#>               plan   pop vap_black_pct reg_black_pct      pp  reock
#>   Enacted BOE district 1 32638      27.87      34.66 0.28895 0.44296
#>   Enacted BOE district 2 32579      28.69      30.34 0.46637 0.45845
#>   Enacted BOE district 3 33090      27.46      27.20 0.39685 0.55962
#>   Enacted BOE district 4 32550      53.73      58.21 0.26578 0.51411
#>   Enacted BOE district 5 32776      25.01      29.71 0.17218 0.39206
#>   Ceiling, 5 districts 32850      67.10      62.78 0.00570 0.05621
#>   Ceiling, 4 districts 41635      62.70      58.82 0.00480 0.07642
#>   Reach 1, 5 districts 32738      41.27      48.17 0.32387 0.76630
#>   Reach 40, 5 districts 32769      51.07      55.57 0.03574 0.33556
#>   Reach unlimited, 5 districts 32731      60.00      58.04 0.00763 0.28433
#>   Reach 1, 4 districts 40910      44.33      50.37 0.36959 0.79365
#>   Reach unlimited, 4 districts 40939      56.53      54.97 0.00672 0.30628
```

```
par(mfrow = c(1, 3), mar = c(0.3, 0.3, 2.0, 0.3))
draw <- function(nm, colr, ttl) {
  plot(NA, xlim = range(bs$x), ylim = range(bs$y), asp = 1, axes = FALSE,
       xlab = "", ylab = "")
  z <- ps[ps$id == "COUNTY", ]
  for (r in split(z, z$part)) polygon(r$x, r$y, col = "#f2f2f2", border = "grey65")
  z <- ps[ps$id == nm, ]
  for (r in split(z, z$part)) polygon(r$x, r$y, col = colr, border = colr)
  s <- d[d$plan == nm, ]
  mtext(ttl, 3, line = 0.85, cex = 0.74)
  mtext(sprintf("%.1f%% Black adults, PP %.3f", s$vap_black_pct, s$pp), 3,
        line = 0.05, cex = 0.64, col = "#666")
}
draw("Reach 1, 5 districts", "#2c7fb8", "Algorithm, reach 1")
draw("Reach unlimited, 5 districts", "#e08214", "Algorithm, reach unlimited")
draw("Enacted BOE district 4", "#C41230", "Enacted District 4")
```



```
cat(sprintf("enacted District 4: %.2f%% Black adults, Polsby-Popper %.3f\n",
            e4$vap_black_pct, e4$pp))
```

```
#> enacted District 4: 53.73% Black adults, Polsby-Popper 0.266
```

```
cat(sprintf("algorithm districts that beat it on BOTH measures: %d of %d\n",
            sum(t5$vap_black_pct >= e4$vap_black_pct & t5$pp >= e4$pp), nrow(t5)))
```

```
#> algorithm districts that beat it on BOTH measures: 0 of 15
```

The algorithm’s failure was a fact about the algorithm. A district that already existed, drawn by hand out of the whole county, dominated everything it produced.

This does **not** show how the enacted plan was drawn, and it is not a comparison of two illustrative maps. What it shows is about evidence. **Exhibiting a map settles the question in the affirmative for good. Failing to find one settles nothing**, because the search was never exhaustive and never could be. The first *Gingles* precondition is provable and not disprovable, and that is exactly where *Callais* intervened.

Part 8 — “Majority” of what?

Bartlett requires a majority. Of what? Take the **four-district** configuration, where the three denominators disagree.

```
t4 <- tr[tr$districts == 4, ]
f4 <- t4[is.finite(t4$reach) & t4$reach <= 100, ]
o <- f4[, c("reach", "pop_black_pct", "vap_black_pct", "reg_black_pct")]
o$majority_on_adults <- f4$vap_black_pct > 50
o$majority_on_registered <- f4$reg_black_pct > 50
print(o, row.names = FALSE, digits = 4)
```

```
#> reach pop_black_pct vap_black_pct reg_black_pct majority_on_adults
#>      1          48.21          44.33          50.37          FALSE
```

```

#>      2      47.92      44.05      50.08      FALSE
#>      3      49.32      45.31      50.57      FALSE
#>      4      49.57      45.57      50.71      FALSE
#>      6      49.25      45.38      50.77      FALSE
#>      8      49.88      45.91      51.03      FALSE
#>     12      50.67      46.78      51.71      FALSE
#>     16      50.77      46.84      51.88      FALSE
#>     24      52.19      48.30      53.20      FALSE
#>     40      52.75      48.93      53.98      FALSE
#>     60      53.06      49.28      54.07      FALSE
#>    100      53.12      49.59      53.73      FALSE
#> majority_on_registered
#>                TRUE
#>                TRUE
#>                TRUE
#>                TRUE
#>                TRUE
#>                TRUE
#>                TRUE
#>                TRUE
#>                TRUE
#>                TRUE
#>                TRUE
#>                TRUE
#>                TRUE

```

Every one of those districts is majority-Black among registered voters. Not one of them is majority-Black among voting-age population. Same blocks, same lines, same people, two answers to the question *Bartlett* says is dispositive.

Courts generally work in voting-age or citizen-voting-age population — the more conservative of the two here. But the choice is a choice, and it decides the question.

Part 9 — Does the starting block matter?

```

sw <- read.csv("data/seed_sweep.csv")
cat(sprintf("starting blocks tried: %d\n", nrow(sw)))

```

```

#> starting blocks tried: 306

```

```

cat(sprintf("results ran %.2f%% to %.2f%%, sd %.2f points\n",
            min(sw$d5_vap_black_pct), max(sw$d5_vap_black_pct),
            sd(sw$d5_vap_black_pct)))

```

```

#> results ran 59.81% to 61.89%, sd 0.48 points

```

```

cat(sprintf("\nfor comparison, the dial moved the answer %.1f points\n",
            max(t5$vap_black_pct) - min(t5$vap_black_pct[t5$reach == 1])))

```

```

#>
#> for comparison, the dial moved the answer 18.7 points

```

Where you start barely matters. How far you are willing to reach matters enormously. Not every analyst choice moves the answer — and the one that does is the one encoding what you think a district is allowed to look like.

Part 10 — Your turn

Change **one** thing, re-run, and write about what changed. Pick one.

Option A — Move the dial yourself. Run `grow(seed, ideal, k)` for a value of `k` not in the table — try 5, 30, or 150 — and `score()` the result. Where does your district sit against the curve in Part 6, and could you defend that value of `k` to a judge who asked why you chose it?

Option B — Start somewhere else. Part 9 says the seed barely matters for the unlimited-reach algorithm. Test whether that holds at `reach = 1`, where race is out of the annexation rule: pick three different seeds (`order(-b$vap_black)[c(1, 10, 50)]`) and `score(grow(s, ideal, 1))` each. Does the answer depend on the starting block more or less than the race-aware version does, and why would that be? Then the harder half: those three seeds were themselves ranked by Black adults. **If race may not be a districting criterion, is `reach = 1` from a race-chosen seed a race-blind map or not?**

Option C — Change the denominator. `grow()` maximises the Black share of **adults**. Rewrite it to maximise the share of **registered voters** instead (swap `b$vap_black/b$vap` for `b$reg_black/b$reg` inside the function). Compare the district you get to the one in Part 5. Which denominator should an expert use, and what is the argument for the other one?

Option D — Four districts instead of five. Set `ideal4 <- sum(b$pop) / 4` and re-run Parts 5 and 6. A smaller board means larger districts. Does that make a majority-Black district easier or harder, and does your answer depend on which denominator you count?

Option E — Your own question. 3,269 blocks, an adjacency list, three populations, and an enacted map to check yourself against.

What this lab cannot tell you

- **Whether anyone’s vote is diluted.** Nothing here touches preconditions two and three, or the totality of the circumstances, which is where these cases are decided. **Precondition one is a threshold, not a verdict.**
- **Whether a *second* majority-Black district could be drawn.** The algorithm builds one district and stops. A real demonstration plan must assign every block and hold all five inside the population band at once — a harder problem that can fail for reasons having nothing to do with the group.
- **Whether any of this would be admissible.** *Callais’s* second condition wants a complete plan that meets all of Georgia’s stated objectives, political ones included, “just as well” as the enacted map. That needs a full partition, a statement of what those objectives were, and election data. This folder has none of the three. **We measure the first condition and are silent on the second**, which is the harder one.
- **What is actually achievable.** Part 7 is the proof: our search failed and a better map existed. The frontier in Part 6 is a lower bound on what a mapmaker could do, and nothing more.
- **Anything about compactness scores as facts.** A district bounded by a winding river scores badly for reasons no mapmaker chose, and one can be made to score better by smoothing a boundary without moving a voter.
- **How people vote.** No election results are in this file. Whether a district *performs* is a separate question.
- And notice what the exercise required: to ask whether Black voters *could* have a district, the algorithm had to sort every block by race. **There is no race-blind way to ask this question.**

On the discussion board

By **Friday, 11:59 p.m.** No length requirement; a sentence or two is enough, so long as it shows you engaged with what we did. Any *one* of the usual four.

Three that would make good posts:

- Part 6 concluded that no compact majority-Black district could be drawn, and Part 7 showed one that already existed. What is the general lesson about evidence produced by a search?
 - The same district is majority-Black on one denominator and not on another. If you were the judge, which would you ask for, and what would you say to the side that wanted the other one?
 - Section 2 remedies require sorting voters by race in order to stop voters being sorted by race, and *Callais* has now said a plaintiff “cannot use race as a districting criterion” in the map that proves the claim. Is there a coherent way out of that, or is the statute asking for something impossible?
-

Where the data came from

2020 Census, P.L. 94-171 redistricting file at block level for Houston County, Georgia: population and voting-age population, both by race. **Black is any-part Black** (alone or in combination), matching Voting Rights Act practice; the surnames lab documents what the alternative definition costs.

2020 TIGER/Line tabulation blocks for Georgia supply geometry, the adjacency list, and both compactness measures. TIGER’s independent POP20 field agrees with the census tables at 163,633 people.

The Houston County Board of Education plan, as enacted by the Georgia General Assembly and in force, as a block-equivalency file.

The Georgia voter registration file for Houston County: **self-reported** race and census block. Rows whose race had been filled in by inference rather than reported by the voter were dropped upstream — see `bisg-check/data/build-data.R`.

Every district here other than the enacted five was drawn by the `grow()` function printed in Part 5. The assembly, including the compactness geometry base R cannot compute, is in `data/build-data.R`.

Houston County’s method of electing its Board of Commissioners is the subject of pending litigation, *Driver v. Houston County*, No. 5:25-cv-00025 (M.D. Ga.), in which your instructor has been retained by plaintiffs. **No material from any expert report is used here, and this lab reaches no legal conclusion about that case or any other.**